Supplementary material

Survivor loss and fuzzy cutvertices: mathematical modelling of key-person risk in informal communication networks

Numbers of sections, definitions, propositions, equations and tables refer to the main article.

S1. Proof of Proposition 3

(1) $\sigma(v) \in [0,1]$ because $0 \le a_v \le a_{\max}$ and $\ln(1+x)$ is increasing. Since $0 \le 1-e^{-\lambda w} < 1$, $\mu(u,v)$ never exceeds $\sigma(u) \wedge \sigma(v)$, and μ is symmetric because $w_{uv} = w_{vu}$. (2) If u and v communicated, then $w_{uv} > 0$ and $a_u, a_v > 0$, so $\sigma(u), \sigma(v) > 0$ and $\mu(u,v) > 0$; otherwise $w_{uv} = 0$ and $\mu(u,v) = 0$. (3) With σ fixed, $1-e^{-\lambda w}$ is strictly increasing in w. □

Part (3) is stated for fixed σ because σ itself depends on the link weights through a_v and $a_{\max}$: raising one w_{uv} can change the memberships of u, v and, through $a_{\max}$, of other members. If the log contains no interactions, $a_{\max} = 0$ and σ is undefined; by convention $\sigma \equiv 0$, so G has no arcs and $Q(G) = \mathrm{NCI}(G) = 0$.

S2. General form of Corollary 1

For a departing set $S \subset V$, the surviving-pair capacity before departure follows by inclusion–exclusion:

$$Q_{\mathrm{surv}}(G;S) = Q(G) - \sum_{w \in S} \sum_{x \neq w} \mathrm{CONN}_G(w,x) + \sum_{\substack{w < w' \\ w,w' \in S}} \mathrm{CONN}_G(w,w').$$

The first sum removes every pair with at least one departing member, counting pairs with both members in S twice; the second adds those back once. $Q(G-S)$ is obtained by applying Proposition 4 to $G-S$.

S3. Algorithm 1 and implementation

Algorithm 1 (fuzzy bridges and hidden key vertices, from Proposition 5).

```
Input : fuzzy graph G = (sigma, mu); set M of all distinct positive
        arc memberships (exact values)
Output: fuzzy bridges B, fuzzy cutvertices C, hidden key vertices H
1  B <- {} ; C <- {}
2  for each t in M (descending):
3      H_t <- crisp graph on V with arcs {(u,v) : mu(u,v) >= t}
4      run Tarjan's bridge / articulation-point search on H_t
5      B <- B + {bridges (u,v) of H_t with mu(u,v) = t}
6      C <- C + {articulation points of H_t}
7  H <- C \ {articulation points of G*}
8  return B, C, H
```

Exact membership values are used: rounding μ creates ties, which add equally strong alternative paths and change which vertices are fuzzy cutvertices. With exact values almost every arc has its own level, so $|M|$ can approach the number of arcs; the following lemma and Proposition 2 reduce the work.

Lemma S1. *Every fuzzy bridge of G belongs to every maximum spanning forest of G^*.*

Proof. Let $e = (u, v)$ be a fuzzy bridge and suppose a maximum spanning forest F does not contain e. By Lemma 1, the u–v path in F has strength $\mathrm{CONN}_G(u, v) \geq \mu(e)$ and avoids e, so $\mathrm{CONN}_{G-e}(u, v) \geq \mu(e)$, contradicting Theorem 1. □

Implementation. (i) Fuzzy bridges are searched only among the at most $n - 1$ arcs of one maximum spanning forest F: an arc $e = (u, v)$ of F is a fuzzy bridge exactly when u and v are disconnected in $H_{\mu(e)} - e$ (Proposition 5(1)). (ii) By Proposition 2, the fuzzy cutvertices are exactly the members with $\mathrm{KS}(w) > 0$, which are computed in any case; the cutvertex output of Algorithm 1 is run as an independent check, and the two sets must coincide. Both shortcuts and Algorithm 1 were verified on 150 random fuzzy graphs (half with tied memberships), where every result matched a brute-force evaluation of the definitions (`verify.py` in the code repository). (iii) Because floating-point arithmetic can leave residual values at machine precision, the implementation treats $\mathrm{KS}(w) > 10^{-12}$ as positive.

S4. Comparison, sensitivity and robustness design

The comparison measures are Kendall's τ between KS and each of S2–S5, the number of shared members in the top-k sets ($k = 10, 20, 50$), and the hidden share (hidden key vertices divided by fuzzy cutvertices, for all and for material ones). Table S1 lists the main and sensitivity settings, all fixed before the analysis. Each membership-parameter setting is compared with the main setting on three criteria: (1) S1 remains first by L_S; (2) Kendall's $\tau \geq 0.70$ between the KS rankings, computed over members with KS > 0 in at least one of the two settings, so the block of zero scores does not inflate agreement; (3) Jaccard similarity $|A \cap B|/|A \cup B| \geq 0.50$ between the top-5% groups, which can differ slightly in size because ties at the cutoff are included. A finding is robust if all three hold in every setting. Table S2 reports the results; Table S3 summarizes the network.

Table S1. Main and sensitivity settings.

Setting	Main value	Sensitivity values
Saturation λ	$\lambda_0 = \ln 2/\mathrm{median}(w_{uv} > 0)$	$0.5\lambda_0$, $2\lambda_0$ (with $\eta = 0$)
Recency η	0 (message counts)	$\eta_0 = 2\ln 2/H$ (with λ_0)
Departure wave K	5% of members	1%, 10%
Materiality proportion p	5% of members by KS	1%, 10%
Random runs	100	none

Table S2. Robustness to the membership parameters, compared with the main setting (λ_0, $\eta = 0$).

Setting	Kendall's τ (KS > 0)	Jaccard, top 5%	S1 rank at 1% / 5% / 10%
$\lambda = 0.5\lambda_0$	0.772	0.613	1 / 1 / 1
$\lambda = 2\lambda_0$	0.815	0.667	1 / 1 / 1
$\eta = \eta_0$ (recency)	0.767	0.724	1 / 1 / 1
Criterion	≥ 0.70	≥ 0.50	first in every wave

Table S3. Network and key-member summary (main setting).

Measure	Value
Members / communicating pairs	986 / 16,064
Distinct membership levels	7,458
Network Cohesion Index	0.374
Fuzzy cutvertices (KS > 0)	270 (27.4% of members)
of which crisp cutvertices	73
of which hidden key members	197 (73.0%)
Material hidden key members, $p = 1\%/5\%/10\%$	4 / 24 / 56
Hidden key members in the top 10 by KS	4
Fuzzy bridges / non-pendant	845 / 750